

VECTORS

- **Scalar Quantity:** - It is a physical quantity which has only magnitude. Examples: - Length, Distance, Speed etc.
- **Vector Quantity:** - It is a physical quantity determined by both magnitude and direction. Example: - Force, Velocity, Displacement. It is always assigned with an arrow or a directed line segment, $\overrightarrow{PQ}$ where P is the initial point and the Q is the terminal point.
- **Length or Magnitude of a Vector:** -It is also called as norm. The magnitude of a vector $\vec{a}$ is denoted by $|\vec{a}|$.
- **Null Vector:** -A vector whose magnitude is zero is called as null vector. Example: - Point. The direction of a null vector is not defined specifically. We denote the null vector by $\vec{0}$.
- **Unit Vector:** -A vector whose magnitude is 1 is called as unit vector. i.e. $|\vec{a}| = 1$.

(a) Unit vector along the direction of $\vec{a}$ is $\frac{\vec{a}}{|\vec{a}|}$.

(b) Unit vector in the opposite direction of $\vec{a}$ is $-\frac{\vec{a}}{|\vec{a}|}$.

(c) Unit vector parallel to $\vec{a}$ is $\pm \frac{\vec{a}}{|\vec{a}|}$.

- **Equality of Two Vectors:** -Two vectors are said to be equal if they have same length and same direction irrespective of their initial and terminal points.
- **Like Vectors:** Vector having the same direction are called like vectors.
- **Component of a Vector:** -Let $\vec{a}$ is in space 3-D
Then components of a vector are the projection of a vector along the co-ordinate axes. If a_1, b_1, c_1 are the projections along x, y, and z axis respectively, then a_1, b_1, c_1 are called as components and can be expressed as $[a_1, b_1, c_1]$. These components are scalar quantity.

Let vector $\overrightarrow{PQ}$ having $P(x_1, y_1, z_1), Q(x_2, y_2, z_2)$.

Then the components of $\overrightarrow{PQ}$ along co-ordinate axes are $x_2 - x_1, y_2 - y_1, z_2 - z_1$.

Magnitude in terms of components of $\overrightarrow{PQ}$ is $|\overrightarrow{PQ}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$.

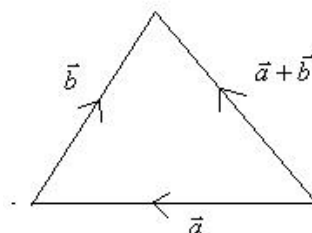
- **Unit vector along the co-ordinate axes:** $\hat{i}, \hat{j}, \hat{k}$ are the vectors along the direction of co-ordinate axes. Where $\hat{i} = [1, 0, 0], \hat{j} = [0, 1, 0], \hat{k} = [0, 0, 1]$ and $|\hat{i}| = |\hat{j}| = |\hat{k}| = 1$.

All the unit vectors $\hat{i}, \hat{j}, \hat{k}$ are perpendicular to each other.

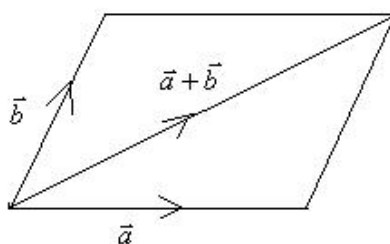
If a vector $\vec{a}$ has components $[p, q, r]$ then $\vec{a} = p\hat{i} + q\hat{j} + r\hat{k}$.

- **Position Vector of a Point:** -Let $P(x, y, z)$ be any point. Then position vector of P be a vector with origin $O(0, 0, 0)$ as the vector with initial point O and terminal point P.
So, $\vec{r} = \overrightarrow{OP} = (x - 0)\hat{i} + (y - 0)\hat{j} + (z - 0)\hat{k} = x\hat{i} + y\hat{j} + z\hat{k}$.

- **Vector Addition:** -Let $\vec{a}$ and $\vec{b}$ are two vectors.
Then their sum is denoted by $\vec{a} + \vec{b}$.
If $\vec{a} = [a_1, b_1, c_1]$, $\vec{b} = [a_2, b_2, c_2]$, then $\vec{a} + \vec{b} = [a_1 + a_2, b_1 + b_2, c_1 + c_2]$
- **Triangle Law of Vector Addition:** - If two vectors $\vec{a}$ and $\vec{b}$ are adjacent sides of a triangle with the same direction then the third side with opposite direction represents $\vec{a} + \vec{b}$.



- **Parallel Law of Vector Addition:** -If two vectors $\vec{a}$ and $\vec{b}$ are represents as two adjacent sides with same initial point. Then resultant vector $\vec{a} + \vec{b}$ is the diagonal with the same initial point.



Properties of Vector addition: -

- $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ Addition is Commutative.
 - $\vec{a} + \vec{0} = \vec{0} + \vec{a} = \vec{a}$ Existence of additive identity.
 - $\vec{a} + (-\vec{a}) = (-\vec{a}) + \vec{a} = \vec{0}$ Existence of additive inverse.
 - $\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$ Addition is Associative.
- **Scalar Multiplication:** - Let $\vec{a} = [a_1, b_1, c_1]$. Then $k\vec{a} = [ka_1, kb_1, kc_1]$ where k is any constant.
 - **Geometrically:** -If $\vec{a} \neq \vec{0}$, then $k\vec{a}$ has the same direction with $\vec{a}$ if $k > 0$ and opposite direction with $\vec{a}$ if $k < 0$ and $k\vec{a} = \vec{0}$ if $k = 0$.

Properties of Scalar Multiplication: -

- $c(\vec{a} + \vec{b}) = c\vec{a} + c\vec{b}$
- $(c + k)\vec{a} = c\vec{a} + k\vec{a}$
- $(ck)\vec{a} = c(k\vec{a})$
- $1\vec{a} = \vec{a}1 = \vec{a}$
- $0\vec{a} = \vec{a}0 = \vec{0}$
- $(-1)\vec{a} = \vec{a}(-1) = -\vec{a}$
- The modulus of $k\vec{a}$ is $k|\vec{a}|$ when $k > 0$ and the direction of $k\vec{a}$ is in the direction of $\vec{a}$.
- The modulus of $k\vec{a}$ is $-k|\vec{a}|$ when $k < 0$ and the direction of $k\vec{a}$ is opposite of $\vec{a}$.

- **Like Vectors and Parallel Vectors:** Vectors which have the same direction are said to be like vectors. Two vectors having the same direction or opposite direction is called as parallel vectors.
- **Like Parallel and Unlike Parallel Vectors:** $\vec{a}$ and $\vec{b}$ are like parallel vectors if they have the same direction. $\vec{a}$ and $\vec{b}$ are unlike parallel vectors if they have opposite direction.
- Let $l \in R$ and $\vec{a}$ be a vector, then
 - (a) If $l > 0$, then $\vec{a}$ and $l\vec{a}$ are like parallel vectors.
 - (b) If $l < 0$, then $\vec{a}$ and $l\vec{a}$ are unlike parallel vectors.
 - (c) If $l = 0$, then $l\vec{a}$ is a null vector.
 - (d) If $l \neq 0$, $a \neq 0$ then $\vec{a}$ and $l\vec{a}$ are collinear or parallel vectors.
- **Angle Between Two Vectors:** If $\vec{OA} = \vec{a}$ and $\vec{OB} = \vec{b}$ be non-zero vectors then $\angle AOB = q$, $0^\circ \leq q \leq 180^\circ$ is defined as the angle between $\vec{a}$ and $\vec{b}$. It is denoted by $(\vec{a}, \vec{b})$.
 - (a) $(\vec{a}, \vec{b}) = 0^\circ \Leftrightarrow \vec{a}$ and $\vec{b}$ are like vectors.
 - (b) $(\vec{a}, \vec{b}) = 90^\circ = \frac{\pi}{2} \Leftrightarrow \vec{a}$ and $\vec{b}$ are orthogonal vectors.
 - (c) $(\vec{a}, \vec{b}) = 180^\circ = \pi \Leftrightarrow \vec{a}$ and $\vec{b}$ are unlike vectors.
 - (d) If $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, then angle between $\vec{a}$ and $\vec{b}$ is undefined.
 - (e) $(\vec{a}, \vec{b}) = (\vec{b}, \vec{a})$.
 - (f) $(\vec{a}, \vec{b}) = (-\vec{a}, -\vec{b})$
 - (g) $(m\vec{a}, n\vec{b}) = (\vec{a}, \vec{b}) \quad \forall m, n > 0$.
 - (h) $(m\vec{a}, n\vec{b}) = 180^\circ - (\vec{a}, \vec{b})$ if m and n are of opposite sign.
- **Collinear Vectors:** Vector lie on the same line or parallel lines are called collinear vector or parallel vectors.
 - (a) $\vec{a}$ and $\vec{b}$ are collinear vectors $\Leftrightarrow \vec{a} = l\vec{b}$, where $l \in R$.
 - (b) $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are collinear vectors $\Leftrightarrow \frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$.
- **Process to show three points A, B and C are Collinear:** (adopt any one of the following)
 - (a) Find the equation of line passing through two points and show that it is satisfied by the third point.
 - (b) Show that there exist scalars not all zero $c_1\vec{a}_1 + c_2\vec{a}_2 + c_3\vec{a}_3 = \vec{0}$ and $c_1 + c_2 + c_3 = 0$.
 - (c) Find $\vec{AB}$ and $\vec{AC}$, express one of them as a scalar multiple of other.
- **Co-planar and non co-planar vectors:** Two or more vectors are said to be co-planar if they lie on the same plane or Vector equivalent to vectors lying in a plane are also called co-planar vectors. Vectors which are not co-planar are called as non co-planar vectors.
- **Linearly Independent and Dependent Vectors:** -Vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3, \dots, \vec{a}_n$ are called as linearly independent if $c_1\vec{a}_1 + c_2\vec{a}_2 + c_3\vec{a}_3 + \dots + c_n\vec{a}_n = \vec{0} \Rightarrow c_1 = c_2 = c_3 = \dots = c_n = 0$. And if all $c_1, c_2, c_3, \dots, c_n$ are not zero then they are called as linearly dependent.
Or Given three vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

Find $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

- If $\Delta \neq 0$, then given vectors are Linearly Independent.
 - If $\Delta = 0$, then given vectors are Linearly Dependent.
 - Any two non-collinear vectors are linearly independent.
 - Any two collinear vectors are linearly dependent.
 - Any set of vectors containing the null vector is linearly dependent.
 - Any three co-planar vectors are linearly dependent.
 - Any four vectors in 3D space are always linearly dependent.
 - If $\vec{a}$ and $\vec{b}$ are two non zero, non collinear vectors such that $x\vec{a} + y\vec{b} = \vec{0}$ then $x = y = 0$.
 - Any vector $\vec{r}$ in the plane generated by two non collinear vectors $\vec{a}$ and $\vec{b}$ can be expressed in the form $\vec{r} = x\vec{a} + y\vec{b}$, $x, y \neq 0$.
- Dimensions:** -the maximum possible linearly independent vectors is called as dimensions.
 - Process to Show 4 points A, B, C and D are Co-planar** (adopt any one of the following):
 - Find $\vec{AB}, \vec{AC}, \vec{AD}$. Show that the determinant of their components is zero.
 - Show that there exist scalars c_1, c_2, c_3, c_4 not all zero $c_1\vec{a}_1 + c_2\vec{a}_2 + c_3\vec{a}_3 + c_4\vec{a}_4 = \vec{0}$ and $c_1 + c_2 + c_3 + c_4 = 0$.
 - Direction Cosines and Direction Ratios of a vector:** Let $\hat{i}, \hat{j}, \hat{k}$ be orthogonal unit vectors in the right handed system and $\vec{r}$ is a vector. If $\vec{a} = (\vec{r}, \hat{i})$, $\vec{b} = (\vec{r}, \hat{j})$ and $\vec{g} = (\vec{r}, \hat{k})$, then $l = \cos \vec{a}, m = \cos \vec{b}, n = \cos \vec{g}$ are called as the direction cosines of the vector $\vec{r}$. The numbers proportional to direction cosines of a given vector, then $a = lk, b = mk, c = nk$ are called as the direction ratios of the vector $\vec{r}$, hence $\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$.
 - If $l = \cos \vec{a}, m = \cos \vec{b}, n = \cos \vec{g}$ are direction cosines of a line, then $l^2 + m^2 + n^2 = 1$.
 - If $\vec{OP} = \vec{r}$ and $P(x, y, z)$ then $x = r \cos \vec{a} = rl, y = r \cos \vec{b} = rm, z = r \cos \vec{g} = rn$.
 - The direction cosines of the vectors $\hat{i}, \hat{j}, \hat{k}$ are $(1, 0, 0), (0, 1, 0), (0, 0, 1)$.
 - $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} = lr\hat{i} + mr\hat{j} + nr\hat{k} = r(l\hat{i} + m\hat{j} + n\hat{k})$, where $r = |\vec{r}|$.
 - $\frac{\vec{r}}{|\vec{r}|} = (l\hat{i} + m\hat{j} + n\hat{k})$. Hence the direction cosine of $\vec{r}$ are coefficient of $\hat{i}, \hat{j}, \hat{k}$ in the unit vector of $\vec{r}$.
 - Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then $r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$.
 - DIVISION FORMULA:**
 - If the position vector of the points A, B with respect to O are $\vec{a}, \vec{b}$. If C divides the line segment $\vec{AB}$ internally in the ratio $m : n$, then position vector of C with respect to O is $\vec{OC} = \frac{m\vec{b} + n\vec{a}}{m + n}$.

- (b) If the position vector of the points A, B with respect to O are $\vec{a}, \vec{b}$. If C divides the line segment $\overline{AB}$ externally in the ratio $m:n$, then position vector of C with respect to O is $\vec{OC} = \frac{m\vec{b} - n\vec{a}}{m - n}$.
- (c) If the position vector of the points A, B with respect to O are $\vec{a}, \vec{b}$. If C bisects the line segment $\overline{AB}$, then position vector of C with respect to O is $\vec{OC} = \frac{\vec{b} + \vec{a}}{2}$.
- (d) Two points which divides a line segment into 3 equal parts are called as trisection points in the ratio 1:2 or 2:1.
- (e) Position vector of centroid G of the triangle with position vector of three vertices are $\vec{a}, \vec{b}, \vec{c}$ is $G = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$ and the incentre is $I = \frac{a\vec{a} + b\vec{b} + c\vec{c}}{a + b + c}$, where a, b and c are the length of the sides of the triangle with usual meaning.
- (f) If G is the centroid of the triangle ABC then, $\vec{GA} + \vec{GB} + \vec{GC} = \vec{0}$.
- (g) If $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are position vectors of the vertices A, B, C and D of a tetrahedron ABCD, then the position vector of its centroid is $G = \frac{\vec{a} + \vec{b} + \vec{c} + \vec{d}}{4}$.
- **Rotation of a Vector About an Axis:** Let $\vec{a} = (a_1, a_2, a_3)$.
 - (a) If the system is rotated about x-axis through an angle α , then new co-ordinate of $\vec{a}$ is $(a_1, a_2 \cos \alpha + a_3 \sin \alpha, -a_2 \sin \alpha + a_3 \cos \alpha)$.
 - (b) If the system is rotated about y-axis through an angle α , then new co-ordinate of $\vec{a}$ is $(-a_3 \sin \alpha + a_1 \cos \alpha, a_2, a_3 \cos \alpha + a_1 \sin \alpha)$.
 - (c) If the system is rotated about z-axis through an angle α , then new co-ordinate of $\vec{a}$ is $(a_1 \cos \alpha + a_2 \sin \alpha, -a_1 \sin \alpha + a_2 \cos \alpha, a_3)$.
 - Let O be the origin, $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ be two vectors. Then the point C with position vector $\vec{OC} = \vec{c} = p\vec{a} + q\vec{b}$ lies:
 - (a) Inside triangle OAB if $p > 0, q > 0, p + q < 1$.
 - (b) Outside triangle OAB, but inside $\angle AOB$ if $p > 0, q > 0, p + q > 1$.
 - (c) Outside triangle OAB, but inside $\angle OAB$ if $p < 0, q > 0, p + q < 1$.
 - (d) Outside triangle OAB, but inside $\angle OBA$ if $p > 0, q < 0, p + q < 1$.
 - If ABCD is a parallelogram such that $\vec{AB} = \vec{a}$, $\vec{BC} = \vec{b}$, then $\vec{AC} = \vec{b} + \vec{a}$, $\vec{BD} = \vec{b} - \vec{a}$.
 - The vector bisecting the angle between $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ is of the form $l(\vec{a} + \vec{b})$.
 - The unit vector bisecting the angle between $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ is of the form $\frac{\vec{a} + \vec{b}}{|\vec{a} + \vec{b}|}$.

• **Vector Equation of Line and corresponding Cartesian form of Line:**

- (a) Vector equation of a line parallel to a vector $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and passes through a point whose position vector is $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ is $\vec{r} = \vec{a} + t\vec{b}$, where $t \in R$. Its Cartesian form is $\frac{x-a_1}{b_1} = \frac{y-a_2}{b_2} = \frac{z-a_3}{b_3}$.
- (b) Vector equation of a line parallel to a vector $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and passes through origin is $\vec{r} = t\vec{b}$, where $t \in R$. Its Cartesian form is $\frac{x}{b_1} = \frac{y}{b_2} = \frac{z}{b_3}$.
- (c) Vector equation of a line passes through two points whose position vectors are $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ is $\vec{r} = (1-t)\vec{a} + t\vec{b}$, where $t \in R$. Its Cartesian form is $\frac{x-a_1}{b_1-a_1} = \frac{y-a_2}{b_2-a_2} = \frac{z-a_3}{b_3-a_3}$ or $\frac{x-b_1}{b_1-a_1} = \frac{y-b_2}{b_2-a_2} = \frac{z-b_3}{b_3-a_3}$.

• **Vector Equation of a Plane:**

- (a) Vector equation of a plane passing through a point whose position vector is $\vec{a}$ and parallel to two non-collinear vectors $\vec{b}, \vec{c}$ is $\vec{r} = \vec{a} + s\vec{b} + t\vec{c}$, where $s, t \in R$.
- (b) Vector equation of a plane passing through origin and parallel to two non-collinear vectors $\vec{b}, \vec{c}$ is $\vec{r} = s\vec{b} + t\vec{c}$, where $s, t \in R$.
- (c) Vector equation of a plane passing through three points whose position vector are $\vec{a}, \vec{b}, \vec{c}$ is $\vec{r} = (1-s-t)\vec{a} + s\vec{b} + t\vec{c}$, where $s, t \in R$.
- (d) Vector equation of a plane passing through two point whose position vector are $\vec{a}, \vec{b}$ and parallel to a vector $\vec{c}$ is $\vec{r} = (1-s)\vec{a} + s\vec{b} + t\vec{c}$, where $s, t \in R$.

Scalar Product or dot product or inner product of Vectors:

- Scalar or dot or inner product of two vectors $\vec{a}$ and $\vec{b}$ is $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos q$, where q is the angle between $\vec{a}$ and $\vec{b}$ and $0 \leq q \leq 180^\circ$.
- If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then $\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$.
- Dot product is a scalar quantity.
- If $\vec{a} \cdot \vec{b} = 0 \Rightarrow \vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $(\vec{a}, \vec{b}) = 90^\circ = \frac{\pi}{2}$.
- $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$ and $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$.
- If $\vec{a} \cdot \vec{b} > 0 \Rightarrow (\vec{a}, \vec{b}) < 90^\circ$.
- If $\vec{a} \cdot \vec{b} < 0 \Rightarrow (\vec{a}, \vec{b}) > 90^\circ$.
- $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$.
- $(m\vec{a} + n\vec{b}) \cdot \vec{c} = m(\vec{a} \cdot \vec{c}) + n(\vec{b} \cdot \vec{c})$.
- $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$.
- $\vec{a} \cdot \vec{a} = 0$ iff $\vec{a} = \vec{0}$.

- $\cos q = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$.
- (a) When $q = 0^\circ$, vector having same direction then $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$.
- (b) When $q = 180^\circ$, vector having opposite direction then $\vec{a} \cdot \vec{b} = -|\vec{a}||\vec{b}|$.
- (c) When $q = 90^\circ$, vector having opposite direction then $\vec{a} \cdot \vec{b} = 0$.
- Any vector $\vec{r}$ can be represented as $\vec{r} = (\vec{r} \cdot \hat{i})\hat{i} + (\vec{r} \cdot \hat{j})\hat{j} + (\vec{r} \cdot \hat{k})\hat{k}$.
- **Orthogonality of Two Vectors:** -Two vectors $\vec{a}$ and $\vec{b}$ are said to be orthogonal to each other if $\vec{a} \cdot \vec{b} = 0$ or $\vec{b} \cdot \vec{a} = 0$.
- **Length in Terms of Inner Product:** - $\vec{a} \cdot \vec{a} = a^2 = |\vec{a}|^2 \Rightarrow |\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}}$ and Angle between two vectors $\cos q = \frac{\vec{a} \cdot \vec{b}}{\sqrt{\vec{a} \cdot \vec{a}} \sqrt{\vec{b} \cdot \vec{b}}}$.
- Scalar Orthogonal projection of $\vec{a}$ on $\vec{b}$ is $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$.
- Vector Orthogonal projection of $\vec{a}$ on $\vec{b}$ is $\frac{(\vec{a} \cdot \vec{b})\vec{b}}{|\vec{b}|^2}$.
- Orthogonal Projection of $\vec{a}$ perpendicular to $\vec{b}$ is $\vec{a} - \frac{(\vec{a} \cdot \vec{b})\vec{b}}{|\vec{b}|^2}$.
- $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = a^2 - b^2 = |\vec{a}|^2 - |\vec{b}|^2$.
- $(\vec{a} + \vec{b})^2 = a^2 + 2\vec{a} \cdot \vec{b} + b^2 = |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2$.
- $(\vec{a} - \vec{b})^2 = a^2 - 2\vec{a} \cdot \vec{b} + b^2 = |\vec{a}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2$.
- $(\vec{a} + \vec{b})^2 + (\vec{a} - \vec{b})^2 = 2(a^2 + b^2) = 2(|\vec{a}|^2 + |\vec{b}|^2)$.
- $(\vec{a} + \vec{b})^2 - (\vec{a} - \vec{b})^2 = 4(\vec{a} \cdot \vec{b})$.
- $(\vec{a} + \vec{b} + \vec{c})^2 = a^2 + b^2 + c^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$.
- $(\vec{a} \cdot \vec{b})^2 \leq a^2 b^2$.
- If $\hat{i}, \hat{j}, \hat{k}$ be orthogonal unit vectors in the right handed system then $\hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1$ and $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{i} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{j} = \hat{k} \cdot \hat{i} = \hat{i} \cdot \hat{k} = 0$.
- Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$. If $\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3 = 0$ then $(\vec{a}, \vec{b}) = 90^\circ = \frac{\pi}{2}$.

- Two vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are parallel if $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$.
- $\vec{a} \cdot \vec{a} > 0$.
- $|\vec{a} \cdot \vec{b}| \leq |\vec{a}||\vec{b}|$ (Schwartz inequality).
- $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$ (Triangle inequality).
- $|\vec{a} - \vec{b}| \geq ||\vec{a}| - |\vec{b}||$.
- $|\vec{a} - \vec{b}| \leq |\vec{a}| + |\vec{b}|$.
- If $\vec{a}$ on $\vec{b}$ are like vectors $\Rightarrow \vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$.
- If $\vec{a}$ on $\vec{b}$ are unlike vectors $\Rightarrow \vec{a} \cdot \vec{b} = -|\vec{a}||\vec{b}|$.
- If $\vec{a}$ on $\vec{b}$ are parallel unit vectors $\Rightarrow \vec{a} \cdot \vec{b} = \pm 1$.
- Let force = $\vec{p}$ and displacement = $\vec{d}$. Then work done = $w = \vec{p} \cdot \vec{d}$.
 - (a) If $q < 90^\circ$ then $w > 0$.
 - (b) If $\vec{p}$ and $\vec{d}$ are orthogonal then $w = 0$.
 - (c) If $q > 90^\circ$ then $w < 0$, work done against the force.

Vector or Cross Product of Vectors:

- Cross product is a vector quantity.
- $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin q \cdot \hat{n}$ where q is the angle between $\vec{a}$ and $\vec{b}$, & $\hat{n}$ is the unit vector perpendicular to $\vec{a}$ and $\vec{b}$.
- If $\vec{a} = [a_1, a_2, a_3]$, $\vec{b} = [b_1, b_2, b_3]$, then:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = [a_2b_3 - a_3b_2 \quad a_3b_1 - a_1b_3 \quad a_1b_2 - a_2b_1]$$
- If $\vec{a}$ or $\vec{b}$ is zero vector then $\vec{a} \times \vec{b} = \vec{0}$.
- If $\vec{a}$ and $\vec{b}$ are collinear then $\vec{a} \times \vec{b} = \vec{0}$.
- If $\vec{a}$ and $\vec{b}$ are parallel then $\vec{a} \times \vec{b} = \vec{0}$.
- $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin q \cdot \hat{n} \leq |\vec{a}||\vec{b}|$.
- $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$.
- $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$
- $|\vec{a} \times \vec{b}| = |\vec{b} \times \vec{a}|$
- $\sin q = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$

- $l\vec{a} \times m\vec{b} = lm(\vec{a} \times \vec{b})$.
- $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$
- $(\vec{a} + \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}$.
- $\hat{i} \times \hat{i} = \vec{0}, \hat{j} \times \hat{j} = \vec{0}, \hat{k} \times \hat{k} = \vec{0}$
- $\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}, \hat{j} \times \hat{i} = -\hat{k}, \hat{k} \times \hat{j} = -\hat{i}, \hat{i} \times \hat{k} = -\hat{j}$
- Unit vector perpendicular to $\vec{a}$ and $\vec{b} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$.

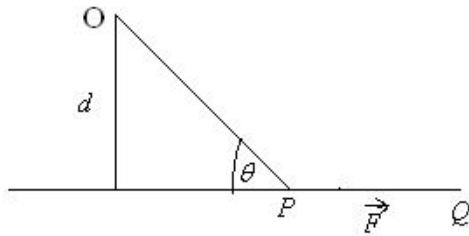
- If $\vec{a} = [a_1, a_2, a_3], \vec{b} = [b_1, b_2, b_3]$, then

$$\text{Then, } \sin \theta = \frac{\sqrt{(a_1b_2 - a_2b_1)^2 + (a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2}}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}}$$

$$\cos \theta = \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}}.$$

- If $\vec{a}$ and $\vec{b}$ are represented by two adjacent sides of a triangle then area of triangle is $\frac{1}{2}|\vec{a} \times \vec{b}|$.
- If $\vec{a}$ and $\vec{b}$ are represented by two adjacent sides of a triangle then vector area of triangle is $\frac{1}{2}(\vec{a} \times \vec{b})$.
- If $A(\vec{a}), B(\vec{b}), C(\vec{c})$ are the vertices of the triangle ABC then vector area is $\frac{1}{2}(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})$.
- If $A(\vec{a}), B(\vec{b}), C(\vec{c})$ are the vertices of the triangle ABC then area is $\frac{1}{2}|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$.
- Let $\vec{a}$ and $\vec{b}$ are represented by two adjacent sides of a parallelogram. Then Area of Parallelogram is $|\vec{a} \times \vec{b}|$.
- Let $\vec{a}$ and $\vec{b}$ are represented by two adjacent sides of a parallelogram. Then vector Area of Parallelogram is $\vec{a} \times \vec{b}$.
- Let $\vec{a}$ and $\vec{b}$ are represented by two diagonals of a parallelogram. Then Area of Parallelogram is $\frac{1}{2}|\vec{a} \times \vec{b}|$.
- Let $\vec{a}$ and $\vec{b}$ are represented by two diagonals of a parallelogram. Then vector Area of Parallelogram is $\frac{1}{2}(\vec{a} \times \vec{b})$.
- If $A(\vec{a}), B(\vec{b}), C(\vec{c})$ points are collinear then $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = \vec{0}$ and vice versa.

- If $\vec{a}$ and $\vec{b}$ are two non zero and non parallel vectors, then $(\vec{a} \times \vec{b})^2 = \vec{a}^2 \vec{b}^2 - (\vec{a} \cdot \vec{b})^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} \end{vmatrix}$.
- If $A(\vec{a})$, $B(\vec{b})$, $C(\vec{c})$ are the vertices of the triangle ABC, then perpendicular distance from C to AB is $\frac{|\vec{AC} \times \vec{AB}|}{|\vec{AB}|} = \frac{|(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})|}{|\vec{b} - \vec{a}|}$.
- The perpendicular distance from a point P to the line joining the points A and B is $\frac{|\vec{AP} \times \vec{AB}|}{|\vec{AB}|}$.
- The length of projection of $\vec{b}$ on a vector $\vec{a}$ is $\frac{|\vec{a} \times \vec{b}|}{|\vec{a}|}$.
- The length of projection of $\vec{a}$ on a vector $\vec{b}$ is $\frac{|\vec{a} \times \vec{b}|}{|\vec{b}|}$.
- For any vector $\vec{a}$, $(\vec{a} \times \hat{i})^2 + (\vec{a} \times \hat{j})^2 + (\vec{a} \times \hat{k})^2 = 2|\vec{a}|^2$.
- **Moment of Force: -**

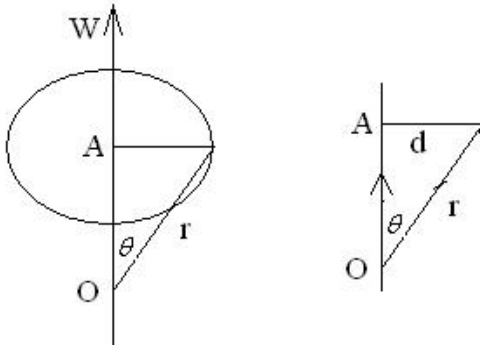


Let $\vec{F}$ is a force act at a point P along a straight line $\overrightarrow{PQ}$. Then moment of force about a point O is given by

$\vec{M}$ = Product of force $\vec{F}$ and perpendicular distance of straight line $\overrightarrow{PQ}$ from O.

$$= |\vec{F}| \cdot d = |\vec{F}| |\vec{OP}| \sin \theta = |\vec{F} \times \vec{OP}| \Rightarrow \vec{M} = \vec{F} \times \vec{OP}.$$

- **Velocity of Rotating Body: -**



Let a rigid body be rotating about OA with angular velocity $\vec{W}$ and its magnitude is $|\vec{W}|$ radian per second and its direction is parallel to axis of rotation OA.

$$\vec{OP} = \vec{r}, \quad \angle AOP = \theta$$

Then Velocity of rotating body = $\vec{W} \times \vec{r}$.

- If $\vec{a}, \vec{b}, \vec{c}$ are triads of right handed or left handed system then $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ are also right handed or left handed system.
- If $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors then $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ are also mutually perpendicular vectors.
- $\vec{a}, \vec{b}, \vec{a} \times \vec{b}$ are right handed triads vectors.
- $\vec{a}, \vec{b}, \vec{b} \times \vec{a}$ are left handed triads vectors.

Product of Three or Four Vectors:

Scalar Triple Product:

- If $\vec{a}, \vec{b}, \vec{c}$ are non zero vectors then scalar triple product is defined as $[\vec{a} \ \vec{b} \ \vec{c}]$ or $\vec{a} \cdot (\vec{b} \times \vec{c})$ or $(\vec{a} \times \vec{b}) \cdot \vec{c}$.
- If $\vec{a} = [a_1, a_2, a_3], \vec{b} = [b_1, b_2, b_3], \vec{c} = [c_1, c_2, c_3]$ then $\vec{a} \cdot (\vec{b} \times \vec{c}) = [\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$.
- $[\vec{a} \ \vec{b} \ \vec{c}] = [\vec{c} \ \vec{a} \ \vec{b}] = [\vec{b} \ \vec{c} \ \vec{a}] = -[\vec{b} \ \vec{a} \ \vec{c}] = -[\vec{a} \ \vec{c} \ \vec{b}] = -[\vec{c} \ \vec{b} \ \vec{a}]$.
- $\vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{c} \times \vec{a}) = \vec{c} \cdot (\vec{a} \times \vec{b}) = (\vec{a} \times \vec{b}) \cdot \vec{c} = (\vec{b} \times \vec{c}) \cdot \vec{a} = (\vec{c} \times \vec{a}) \cdot \vec{b}$.
- $[k\vec{a} \ \vec{b} \ \vec{c}] = k[\vec{a} \ \vec{b} \ \vec{c}]$
- $[p\vec{a} \ q\vec{b} \ r\vec{c}] = pqr[\vec{a} \ \vec{b} \ \vec{c}]$, where p, q, r are scalars.
- $[\hat{i} \ \hat{j} \ \hat{k}] = 1$.
- Scalar triple product of three vectors is zero if any two of them are equal vectors.
- Scalar triple product of three vectors is zero if any two of them are parallel vectors.
- If out of three vectors $\vec{a}, \vec{b}, \vec{c}$ two are equal or parallel then, $[\vec{a} \ \vec{b} \ \vec{c}] = 0$.
- If three vectors $\vec{a}, \vec{b}, \vec{c}$ are coplanar then $[\vec{a} \ \vec{b} \ \vec{c}] = 0$.
- If $[\vec{a} \ \vec{b} \ \vec{c}] \neq 0$, then three vectors $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar.
- If $[\vec{a} \ \vec{b} \ \vec{c}] > 0$, then the vectors $\vec{a}, \vec{b}, \vec{c}$ triad of right handed system.
- If $[\vec{a} \ \vec{b} \ \vec{c}] < 0$, then the vectors $\vec{a}, \vec{b}, \vec{c}$ triad of left handed system.
- The absolute scalar triple product is called as volume of parallelepiped. Let $\vec{a}, \vec{b}, \vec{c}$ are three edge vectors of a parallelogram. Then its volume is given by $|\vec{a} \cdot (\vec{b} \times \vec{c})|$ cu units.

- If three edges of a tetrahedron with same initial points are $\vec{a}, \vec{b}, \vec{c}$, then volume of the tetrahedron is $\frac{1}{6}|\vec{a} \cdot (\vec{b} \times \vec{c})|$ cu units.
- If $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are the vertices of the tetrahedron, then volume of tetrahedron is given by $\frac{1}{6}|\begin{vmatrix} \vec{a}-\vec{b} & \vec{a}-\vec{c} & \vec{a}-\vec{d} \end{vmatrix}|$ cu units.
- If $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are coplanar vectors then $[\vec{a} \ \vec{b} \ \vec{c}] = [\vec{b} \ \vec{c} \ \vec{d}] + [\vec{c} \ \vec{a} \ \vec{d}] + [\vec{a} \ \vec{b} \ \vec{d}]$.
- If three vectors $\vec{a}, \vec{b}, \vec{c}$ are coplanar then $[\vec{a} \times \vec{b} \ \vec{b} \times \vec{c} \ \vec{c} \times \vec{a}] = 0$.
- $[\vec{a} + \vec{b} \ \vec{b} + \vec{c} \ \vec{c} + \vec{a}] = 2[\vec{a} \ \vec{b} \ \vec{c}]$
- If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ then $\vec{a}, \vec{b}, \vec{c}$ are coplanar.
- If $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular unit vectors then $[\vec{a} \ \vec{b} \ \vec{c}] = \pm 1$.
- Three vectors $\vec{a}, \vec{b}, \vec{c}$ are linearly independent if $\vec{a} \cdot (\vec{b} \times \vec{c}) \neq 0$.

Vector Triple Product:

- For three vectors $\vec{a}, \vec{b}, \vec{c}$, their cross product is given by $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$.
- $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$.
- $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$ if $\vec{a}, \vec{c}$ are collinear.
- $\vec{a} \times (\vec{b} \times \vec{c}) = -(\vec{b} \times \vec{c}) \times \vec{a}$.
- $2\vec{r} = \hat{i} \times (\vec{r} \times \hat{i}) + \hat{j} \times (\vec{r} \times \hat{j}) + \hat{k} \times (\vec{r} \times \hat{k})$.
- If $\vec{a}, \vec{b}, \vec{c}$ are non zero vectors and $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{0}$, then $\vec{b}, \vec{c}$ are parallel or collinear vectors.
- If $\vec{a}, \vec{b}, \vec{c}$ are non zero and non parallel vectors then $\vec{a} \times (\vec{b} \times \vec{c}), \vec{b} \times (\vec{c} \times \vec{a}), \vec{c} \times (\vec{a} \times \vec{b})$ are non collinear vectors.
- If $\vec{a}, \vec{b}, \vec{c}$ are vectors then $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times (\vec{a} \times \vec{b}) = \vec{0}$.
- If $\vec{a}, \vec{b}, \vec{c}$ are vectors then $\vec{a} \times (\vec{b} \times \vec{c}), \vec{b} \times (\vec{c} \times \vec{a}), \vec{c} \times (\vec{a} \times \vec{b})$ are coplanar vectors.
- $\vec{a} \times (\vec{b} \times \vec{c})$ is vector lies in the plane $\vec{b}, \vec{c}$ or parallel to the plane $\vec{b}, \vec{c}$.

Dot Product of Four Vectors:

- The dot product of four vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ is defined as :

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = \begin{vmatrix} \vec{a} \cdot \vec{c} & \vec{a} \cdot \vec{d} \\ \vec{b} \cdot \vec{c} & \vec{b} \cdot \vec{d} \end{vmatrix} = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c})$$

Cross Product of Four Vectors:

- The cross product of four vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ is defined as :

$$\begin{aligned} (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) &= [\vec{a} \ \vec{b} \ \vec{d}]\vec{c} - [\vec{a} \ \vec{b} \ \vec{c}]\vec{d} \quad \text{OR} \\ (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) &= [\vec{a} \ \vec{c} \ \vec{d}]\vec{b} - [\vec{b} \ \vec{c} \ \vec{d}]\vec{a} \end{aligned}$$

- $[\vec{b} \ \vec{c} \ \vec{d}]\vec{a} - [\vec{a} \ \vec{c} \ \vec{d}]\vec{b} + [\vec{a} \ \vec{b} \ \vec{d}]\vec{c} - [\vec{a} \ \vec{b} \ \vec{c}]\vec{d} = \vec{0}$,
- Any vector $\vec{r}$ can be expressed as: $\vec{r} = \frac{[\vec{r} \ \vec{b} \ \vec{c}]\vec{a} + [\vec{r} \ \vec{c} \ \vec{a}]\vec{b} + [\vec{r} \ \vec{a} \ \vec{b}]\vec{c}}{[\vec{a} \ \vec{b} \ \vec{c}]}$.

Reciprocal system of vectors:

- If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors then the triads $\vec{a}^{-1}, \vec{b}^{-1}, \vec{c}^{-1}$ given by $\vec{a}^{-1} = \frac{\vec{b} \times \vec{c}}{[\vec{a} \ \vec{b} \ \vec{c}]}$, $\vec{b}^{-1} = \frac{\vec{c} \times \vec{a}}{[\vec{a} \ \vec{b} \ \vec{c}]}$, $\vec{c}^{-1} = \frac{\vec{a} \times \vec{b}}{[\vec{a} \ \vec{b} \ \vec{c}]}$ are called as reciprocal system of vector triads $\vec{a}, \vec{b}, \vec{c}$.
- If $\vec{a}^{-1}, \vec{b}^{-1}, \vec{c}^{-1}$ are reciprocal system of $\vec{a}, \vec{b}, \vec{c}$ then $\vec{a} \cdot \vec{a}^{-1} = \vec{b} \cdot \vec{b}^{-1} = \vec{c} \cdot \vec{c}^{-1} = 1$.
- If $\vec{a}^{-1}, \vec{b}^{-1}, \vec{c}^{-1}$ are reciprocal system of $\vec{a}, \vec{b}, \vec{c}$ then $\vec{a} \cdot \vec{b}^{-1} = \vec{b} \cdot \vec{c}^{-1} = \vec{c} \cdot \vec{a}^{-1} = 0$ and $\vec{a}^{-1} \cdot \vec{b} = \vec{b}^{-1} \cdot \vec{c} = \vec{c}^{-1} \cdot \vec{a} = 0$.
- $[\vec{a}^{-1} \ \vec{b}^{-1} \ \vec{c}^{-1}] = \frac{1}{[\vec{a} \ \vec{b} \ \vec{c}]}$.
- If $\vec{a}, \vec{b}, \vec{c}$ are coplanar vectors, then $\vec{a}^{-1}, \vec{b}^{-1}, \vec{c}^{-1}$ are also coplanar vectors.
- $\hat{i}^{-1} = \hat{i}$, $\hat{j}^{-1} = \hat{j}$, $\hat{k}^{-1} = \hat{k}$.
- If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar vectors for which $[\vec{a} \ \vec{b} \ \vec{c}] \neq 0$ and $\vec{a}^{-1}, \vec{b}^{-1}, \vec{c}^{-1}$ constitute the reciprocal system of vectors, then any vector $\vec{r}$ can be expressed as:
 $\vec{r} = (\vec{r} \cdot \vec{a}^{-1})\vec{a} + (\vec{r} \cdot \vec{b}^{-1})\vec{b} + (\vec{r} \cdot \vec{c}^{-1})\vec{c} = (\vec{r} \cdot \vec{a})\vec{a}^{-1} + (\vec{r} \cdot \vec{b})\vec{b}^{-1} + (\vec{r} \cdot \vec{c})\vec{c}^{-1}$.
- **Some Important Facts:**

(1)

- $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}| \Rightarrow \vec{a} \perp \vec{b}$ i.e. $\vec{a}, \vec{b}$ are orthogonal vectors.
- $|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}| \Rightarrow \vec{a} \parallel \vec{b} \Rightarrow \vec{a} = \lambda \vec{b}$.
- $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 \Rightarrow \vec{a} \perp \vec{b}$ i.e. $\vec{a}, \vec{b}$ are orthogonal vectors.

(2) If $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular i.e. $\vec{a} \perp \vec{b} \perp \vec{c}$ and equal in magnitudes, then $\vec{a} + \vec{b} + \vec{c}$ is equally inclined with $\vec{a}, \vec{b}, \vec{c}$ and angle of inclination with every vector is given by $\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$.

(3) If $\vec{a}, \vec{b}, \vec{c}$ are three vectors, then

- $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times (\vec{c} + \vec{a}) + \vec{c} \times (\vec{a} + \vec{b}) = \vec{0}$.
- $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times (\vec{a} \times \vec{b}) = \vec{0}$.

(4) Three vectors $\vec{a}, \vec{b}, \vec{c}$ are coplanar if and only if:

- $\vec{a} + \vec{b}$, $\vec{b} + \vec{c}$, $\vec{c} + \vec{a}$ are coplanar.
- $\vec{a} \times \vec{b}$, $\vec{b} \times \vec{c}$, $\vec{c} \times \vec{a}$ are coplanar.

- (5) If D, E and F are mid points of the sides BC, CA and AB of a triangle then $\vec{AD} + \vec{BE} + \vec{CF} = \vec{0}$.
- (6) If $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of an equilateral triangle whose orthocentre is at origin then $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ as in equilateral triangle circumcentre, orthocenter, incentre and centroid are coincides with each other.
- (7) **Incentre:** If I is the centre of the circle inscribed in a triangle ABC then value of $a\vec{IA} + b\vec{IB} + c\vec{IC} = \vec{0}$, where $BC = a, CA = b, AB = c$, where $I = \frac{a\vec{a} + b\vec{b} + c\vec{c}}{a + b + c}$
- $$\Rightarrow I = \frac{|\vec{b} - \vec{c}|\vec{a} + |\vec{c} - \vec{a}|\vec{b} + |\vec{a} - \vec{b}|\vec{c}}{|\vec{b} - \vec{c}| + |\vec{c} - \vec{a}| + |\vec{a} - \vec{b}|}.$$
- (8) **Orthocentre:** The position vector of the orthocentre of $\triangle ABC$ is $\frac{\vec{a} \tan A + \vec{b} \tan B + \vec{c} \tan C}{\tan A + \tan B + \tan C}$.
- (9) If ABCD is a rhombus whose diagonals intersects at the origin then $\vec{OA} + \vec{OB} + \vec{OC} + \vec{OD} = \vec{0}$.
- (10) If ABCD is a parallelogram whose diagonals intersects at P if O is the origin then $\vec{OA} + \vec{OB} + \vec{OC} + \vec{OD} = 4\vec{OP}$.
- (11) If $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of three points respectively then $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$ is perpendicular to the plane of ABC.
- (12)
- $[\vec{a} + \vec{b} \quad \vec{b} + \vec{c} \quad \vec{c} + \vec{a}] = 2[\vec{a} \quad \vec{b} \quad \vec{c}].$
 - $[\vec{a} - \vec{b} \quad \vec{b} - \vec{c} \quad \vec{c} - \vec{a}] = 0.$
 - $[\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}] = [\vec{a} \quad \vec{b} \quad \vec{c}]^2.$
 - $[\vec{a} \quad \vec{b} \quad \vec{c}][\vec{p} \quad \vec{q} \quad \vec{r}] = \begin{vmatrix} \vec{a} \cdot \vec{p} & \vec{a} \cdot \vec{q} & \vec{a} \cdot \vec{r} \\ \vec{b} \cdot \vec{p} & \vec{b} \cdot \vec{q} & \vec{b} \cdot \vec{r} \\ \vec{c} \cdot \vec{p} & \vec{c} \cdot \vec{q} & \vec{c} \cdot \vec{r} \end{vmatrix}.$
 - $\vec{r}[\vec{a} \quad \vec{b} \quad \vec{c}] = [\vec{r} \quad \vec{b} \quad \vec{c}]\vec{a} + [\vec{r} \quad \vec{c} \quad \vec{a}]\vec{b} + [\vec{r} \quad \vec{a} \quad \vec{b}]\vec{c}.$
 - $\vec{r}[\vec{a} \quad \vec{b} \quad \vec{c}] = (\vec{r} \cdot \vec{a})(\vec{b} \times \vec{c}) + (\vec{r} \cdot \vec{b})(\vec{c} \times \vec{a}) + (\vec{r} \cdot \vec{c})(\vec{a} \times \vec{b}).$
 - $(\vec{a} \times \vec{b})[\vec{p} \quad \vec{q} \quad \vec{r}] = \begin{vmatrix} \vec{p} & \vec{q} & \vec{r} \\ \vec{a} \cdot \vec{p} & \vec{a} \cdot \vec{q} & \vec{a} \cdot \vec{r} \\ \vec{b} \cdot \vec{p} & \vec{b} \cdot \vec{q} & \vec{b} \cdot \vec{r} \end{vmatrix}.$
- (13) **Perpendicular distance of a point from a line:** Let L is the foot of the perpendicular drawn from $P(\vec{a})$ on the line $\vec{r} = \vec{a} + \lambda \vec{b}$. Since $\vec{r}$ denotes the position vector of any point on the line $\vec{r} = \vec{a} + \lambda \vec{b}$.
 Position Vector of L =
 Position Vector of $\vec{PL} =$

Length PL is the magnitude of $\overrightarrow{PL}$ and is the perpendicular distance of a point from a line.

- (14) **Image of a point in a straight line:** Let $Q(\vec{b})$ is the image of $P(\vec{a})$ in line $\vec{r} = \vec{a} + t\vec{b}$, then

$$\text{Position Vector of } Q = \vec{b} = 2\vec{a} - \vec{a} - \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b}.$$

$$\text{Position vector of } \overrightarrow{PQ} = 2\vec{a} - 2\vec{a} - \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b}.$$

- (15) **Shortest Distance of two Lines:** Let $\vec{r} = \vec{a}_1 + t\vec{b}_1$ and $\vec{r} = \vec{a}_2 + m\vec{b}_2$ are two lines,

$$\text{then shortest distance between them is: } \frac{|(\vec{b}_1 \cdot \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}_1 \times \vec{b}_2|} = \frac{|\vec{b}_1 \cdot \vec{b}_2 \cdot \vec{a}_2 - \vec{a}_1|}{|\vec{b}_1 \times \vec{b}_2|}.$$

- (16) **Shortest Distance of two Parallel Lines:** Let $\vec{r} = \vec{a}_1 + t\vec{b}$ and $\vec{r} = \vec{a}_2 + m\vec{b}$ are two

$$\text{lines, then shortest distance between them is: } \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}.$$

- (17) The perpendicular distance of a point having position vector $\vec{a}$ from the plane $\vec{r} \cdot \vec{n} = d$ is given by $p = \frac{|\vec{a} \cdot \vec{n} - d|}{|\vec{n}|}$.

- (18) Perpendicular distance of a point $P(\vec{r})$ from a plane passing through three points $\vec{a}, \vec{b}, \vec{c}$ is given by $\frac{(\vec{r} - \vec{a}) \cdot (\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b})}{|\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b}|}$.

- (19) Angle between a line $\vec{r} = \vec{a} + t\vec{b}$ and plane $\vec{r} \cdot \vec{n} = d$ is $\theta = \sin^{-1} \left(\frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|} \right)$.

- (20) The equation of sphere with centre at $C(\vec{c})$ and radius a is $|\vec{r} - \vec{c}| = a$.

- (21) **Vector Equation of Plane:**

- Vector equation of plane passing through three non collinear points having position vectors $\vec{a}, \vec{b}, \vec{c}$ is $[\vec{r} - \vec{a} \quad \vec{r} - \vec{b} \quad \vec{r} - \vec{c}] = 0$.
- Vector equation of plane passing through three non collinear points having position vectors $\vec{a}, \vec{b}, \vec{c}$ is $\vec{r} \cdot (\vec{b} \times \vec{c} + \vec{c} \times \vec{a} + \vec{a} \times \vec{b}) = [\vec{a} \quad \vec{b} \quad \vec{c}]$.
- Vector equation of plane passing through a point $\vec{a}$ and parallel to $\vec{b}, \vec{c}$ is $[\vec{r} - \vec{a} \quad \vec{b} \quad \vec{c}] = 0$ or $[\vec{r} \quad \vec{b} \quad \vec{c}] = [\vec{a} \quad \vec{b} \quad \vec{c}]$.

- (d) Vector equation of plane passing through two point $\vec{a}, \vec{b}$ and parallel to $\vec{c}$ is $[\vec{r} - \vec{a} \quad \vec{b} - \vec{a} \quad \vec{c}] = 0$.
- (e) Vector equation of plane through the intersection of planes $\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$ is $\vec{r} \cdot (\vec{n}_1 + \lambda \vec{n}_2) = d_1 + \lambda d_2$.
- (f) Length of perpendicular from origin to a plane containing three points $\vec{a}, \vec{b}, \vec{c}$ is
$$\frac{|\vec{a} \quad \vec{b} \quad \vec{c}|}{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}.$$
- (g) Vector equation of plane passing through origin and $\vec{b}, \vec{c}$ is $[\vec{r} \quad \vec{b} \quad \vec{c}] = 0$.
- (h) The shortest distance between the skew lines $\vec{r} = \vec{a} + t\vec{b}$ and $\vec{r} = \vec{c} + s\vec{d}$ is
$$\frac{|\vec{a} - \vec{c} \quad \vec{b} \quad \vec{d}|}{|\vec{b} \times \vec{d}|}.$$

(22) Vector Equation of Plane in Normal Form – Angle between two Planes:

- (a) Equation of plane at distance p ($p > 0$) from the origin O and perpendicular to $\hat{n}$ is $\vec{r} \cdot \hat{n} = p$. If direction cosine of $\hat{n}$ is l, m, n then Cartesian equation of a plane in normal form is $lx + my + nz = p$.
- (b) Equation of plane through origin O and perpendicular to $\hat{n}$ is $\vec{r} \cdot \hat{n} = 0$. If direction cosine of $\hat{n}$ is l, m, n then Cartesian equation of a plane in normal form is $lx + my + nz = 0$.
- (c) Vector equation of plane passing through origin and perpendicular to $\vec{b}$ is $\vec{r} \cdot \vec{b} = 0$.
- (d) Vector equation of plane passing through a point with position vector $\vec{a}$ and perpendicular to $\hat{n}$ is $(\vec{r} - \vec{a}) \cdot \hat{n} = 0$. Perpendicular distance from origin to this plane is $|\vec{a} \cdot \hat{n}|$.
- (e) Vector equation of plane passing through a point with position vector $\vec{a}$ and perpendicular to $\vec{b}$ is $(\vec{r} - \vec{a}) \cdot \vec{b} = 0$. Perpendicular distance from origin to this plane is
$$\frac{|\vec{a} \cdot \vec{b}|}{|\vec{b}|}.$$
- (f) The Cartesian equation of plane passing through the point $A(x_1, y_1, z_1)$ and perpendicular to the vector $a\hat{i} + b\hat{j} + c\hat{k}$ is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$.
- (g) If $\vec{r} \cdot \vec{m}_1 = 0$ and $\vec{r} \cdot \vec{m}_2 = 0$ are two planes, then angle between these two planes is
$$\cos^{-1} \left(\frac{\vec{m}_1 \cdot \vec{m}_2}{|\vec{m}_1| |\vec{m}_2|} \right).$$

(23) **Vector Equation of Sphere:**

- (a) Vector equation of sphere whose centre has position vector $\vec{c}$ and radius a is $|\vec{r} - \vec{c}| = a$ or $\vec{r}^2 - 2\vec{r} \cdot \vec{c} + (\vec{c}^2 - a^2) = 0$.
- (b) If centre is (a, b, g) and radius a then Cartesian equation of sphere is $(x - a)^2 + (y - b)^2 + (z - g)^2 = a^2$.
- (c) If $A(\vec{a}), B(\vec{b})$ are ends of diameter of a sphere then sphere equation is $(\vec{r} - \vec{a}) \cdot (\vec{r} - \vec{b}) = 0$ or $\vec{r}^2 - 2\vec{r} \cdot (\vec{a} + \vec{b}) + \vec{a} \cdot \vec{b} = 0$.
